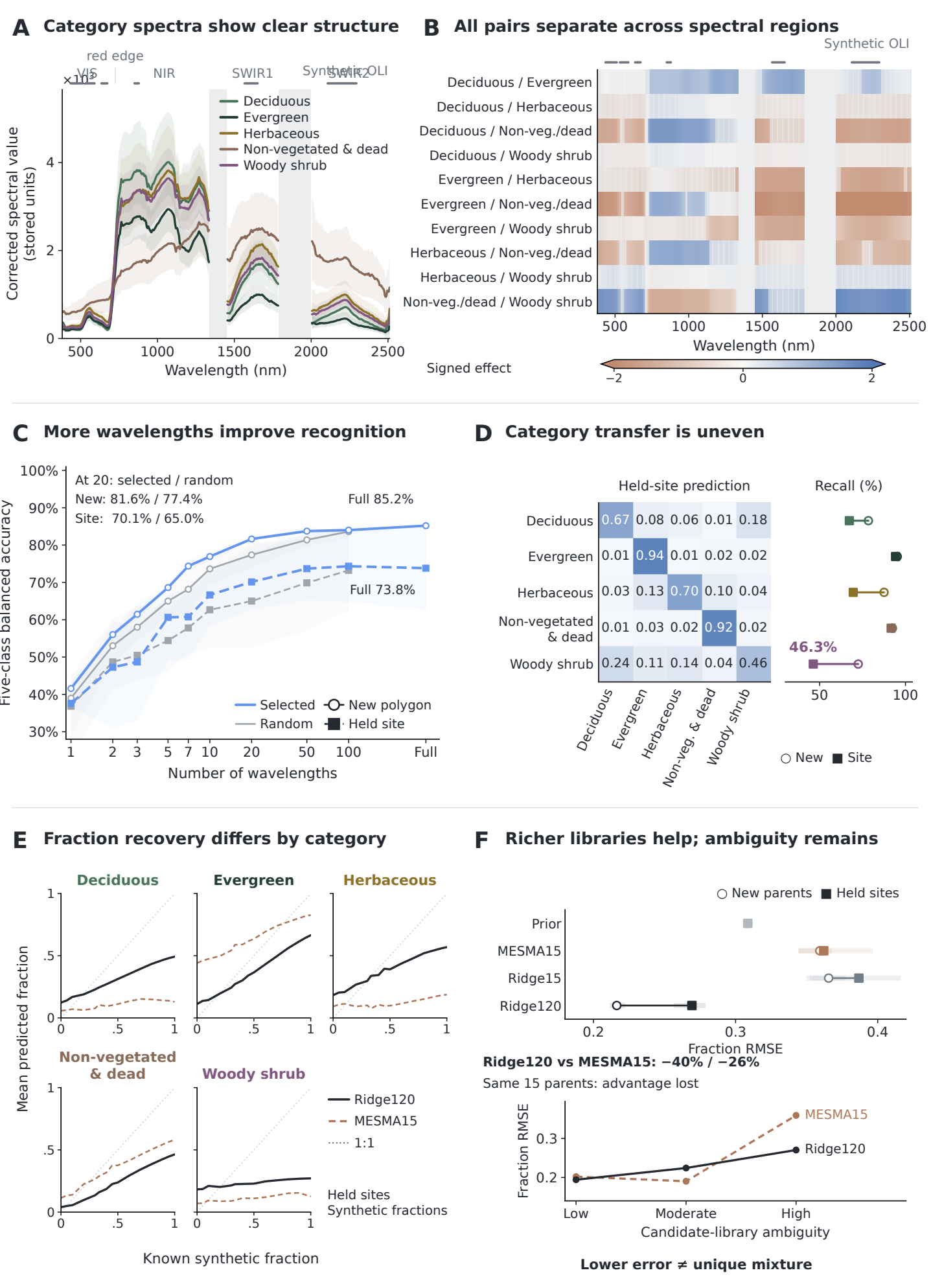

A
Category spectra show clear structure
red edge
VIS
NIR
SWIR1
Synthetic OLI
SWIR2
×10³
Deciduous
Evergreen
Herbaceous
Non-vegetated & dead
Woody shrub
Corrected spectral value (stored units)
Wavelength (nm)
B
All pairs separate across spectral regions
Synthetic OLI
Deciduous / Evergreen
Deciduous / Herbaceous
Deciduous / Non-veg./dead
Deciduous / Woody shrub
Evergreen / Herbaceous
Evergreen / Non-veg./dead
Evergreen / Woody shrub
Herbaceous / Non-veg./dead
Herbaceous / Woody shrub
Non-veg./dead / Woody shrub
Wavelength (nm)
Signed effect
C
More wavelengths improve recognition
At 20: selected / random
New: 81.6% / 77.4%
Site: 70.1% / 65.0%
Full 85.2%
Full 73.8%
Five-class balanced accuracy
Selected
Random
New polygon
Held site
Number of wavelengths
D
Category transfer is uneven
Held-site prediction
Recall (%)
Deciduous 0.67 0.08 0.06 0.01 0.18
Evergreen 0.01 0.94 0.01 0.02 0.02
Herbaceous 0.03 0.13 0.70 0.10 0.04
Non-vegetated & dead 0.01 0.03 0.02 0.92 0.02
Woody shrub 0.24 0.11 0.14 0.04 0.46
46.3%
Deciduous
Evergreen
Herbaceous
Non-veg. & dead
Woody shrub
New
Site
E
Fraction recovery differs by category
Deciduous
Evergreen
Herbaceous
Non-vegetated & dead
Woody shrub
Mean predicted fraction
Known synthetic fraction
Ridge120
MESMA15
1:1
Held sites
Synthetic fractions
F
Richer libraries help; ambiguity remains
New parents
Held sites
Prior
MESMA15
Ridge15
Ridge120
Fraction RMSE
Ridge120 vs MESMA15: −40% / −26%
Same 15 parents: advantage lost
Fraction RMSE
MESMA15
Ridge120
Low
Moderate
High
Candidate-library ambiguity
Lower error ≠ unique mixture